

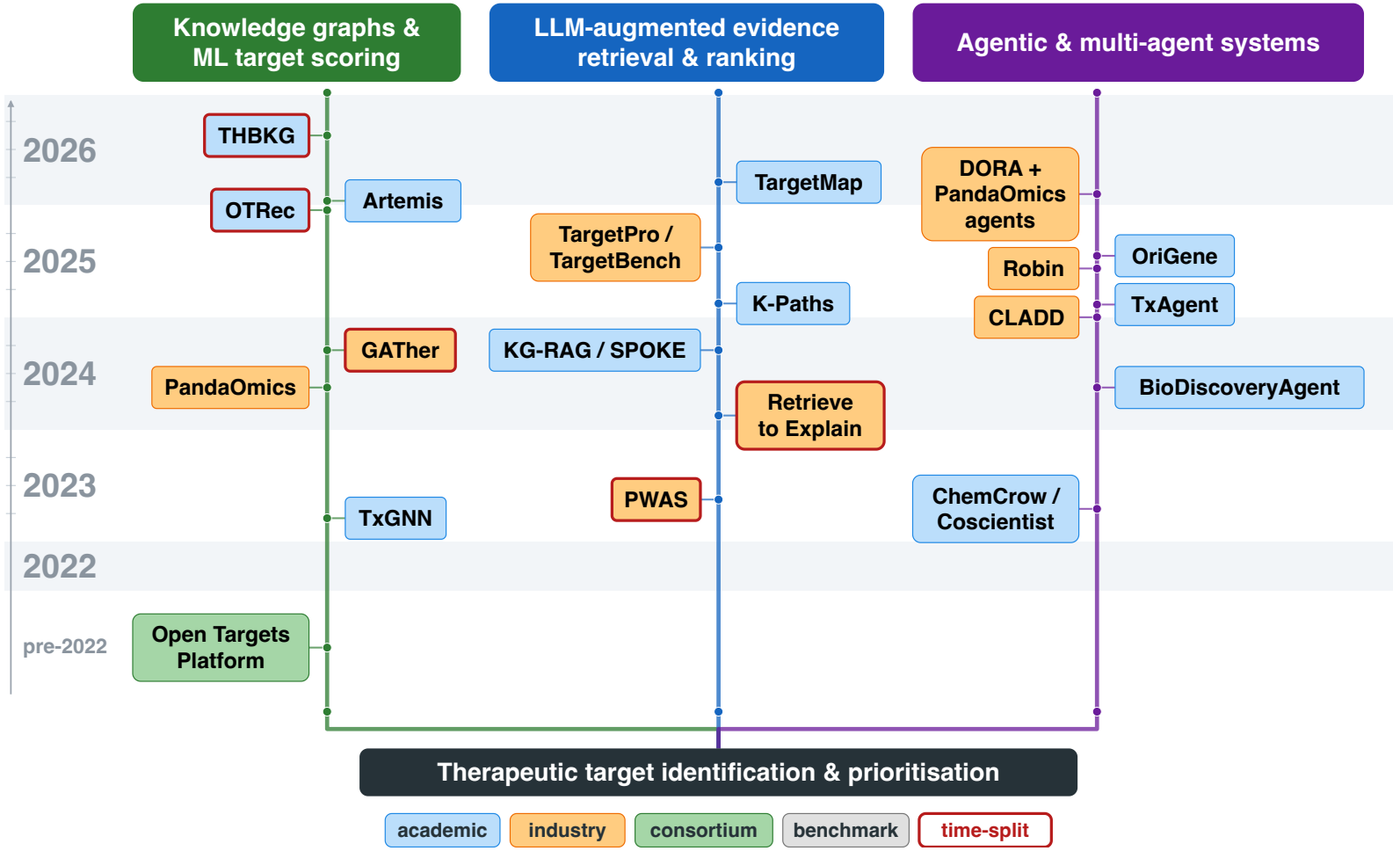
Did the Model Forecast It, or Already Know It? Evaluating Large Language Models for Novel Drug-Target Prioritisation with a Temporal Biomedical Knowledge-Graph Oracle

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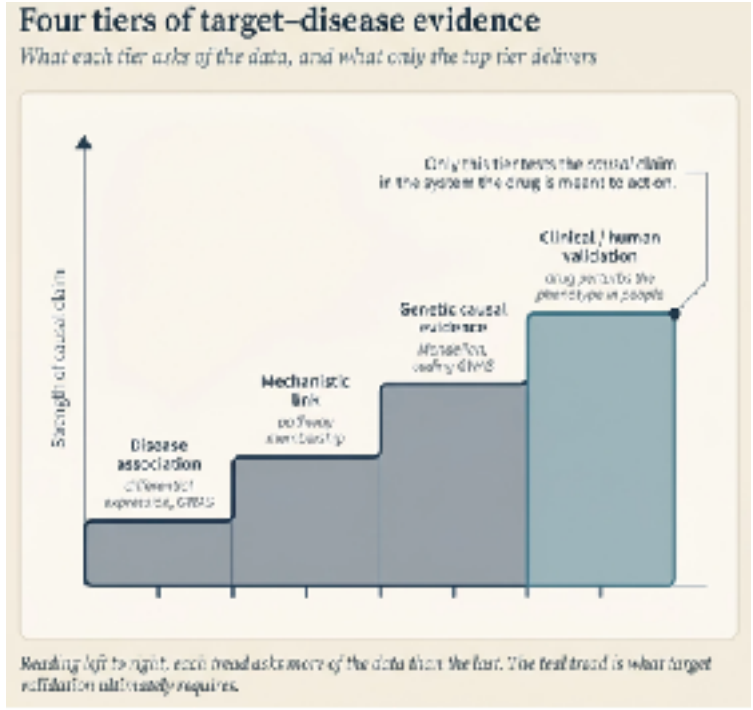
Queen Mary University of London Recursion AIDD CTP

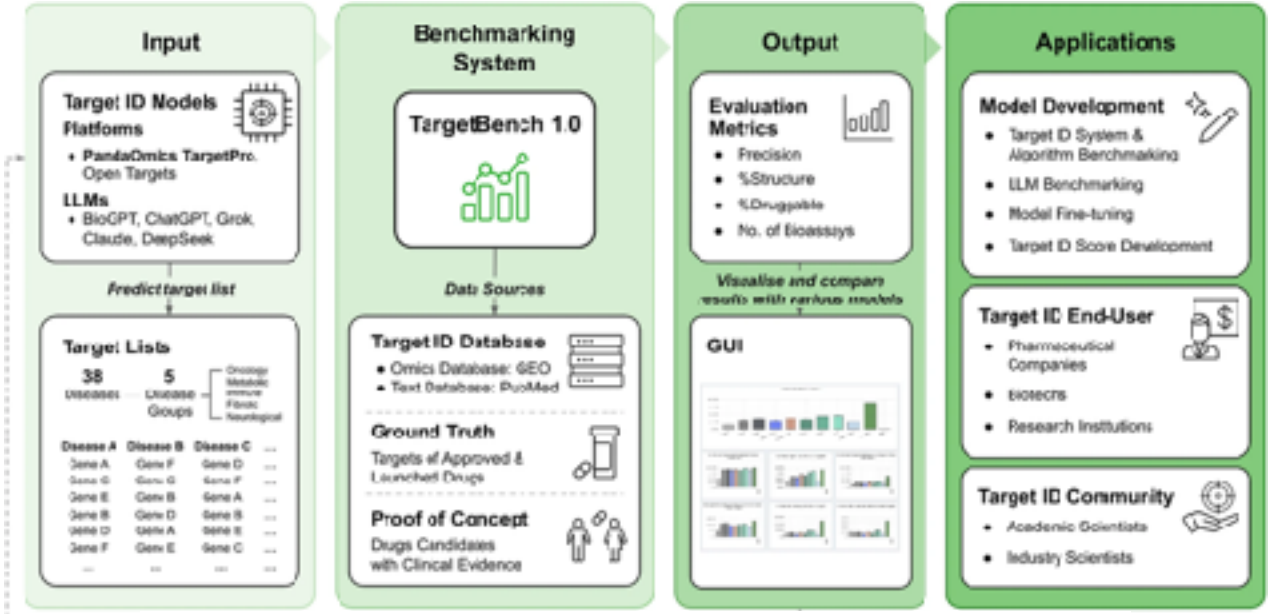
Target discovery platforms, LLMs included, are benchmarked on how many already-known targets they recover

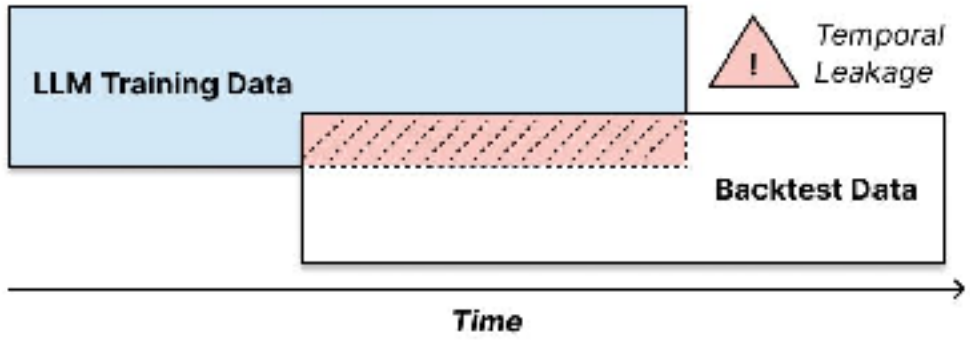


Systems placed at first public appearance. Red border = the paper imposes a literature cutoff or time split: PWAS (Narganes-Carlón et al. 2023), Retrieve to Explain (Patel et al. 2024), GATher (Narganes-Carlón et al. 2024), OTRec (Ofer & Linial 2025), THBKG (Siu et al. 2026). TargetBench: Leung et al., Sci. Rep. 2026.

The evaluation corpus overlaps with the corpus the ML methods were trained on, in both features and labels







Left: N. Flynn, "Target–disease association evidence", noahrflynn.com/blog/2026. Right: TargetBench 1.0 overview, Leung et al., Sci. Rep. 2026.

LLMs reason well about mechanisms; whether they forecast novel target–disease pairs or recall known ones is untested

- **The bottleneck is disease understanding**
  Identifying a mechanism whose modification actually changes the course of disease in patients (Koller 2026).

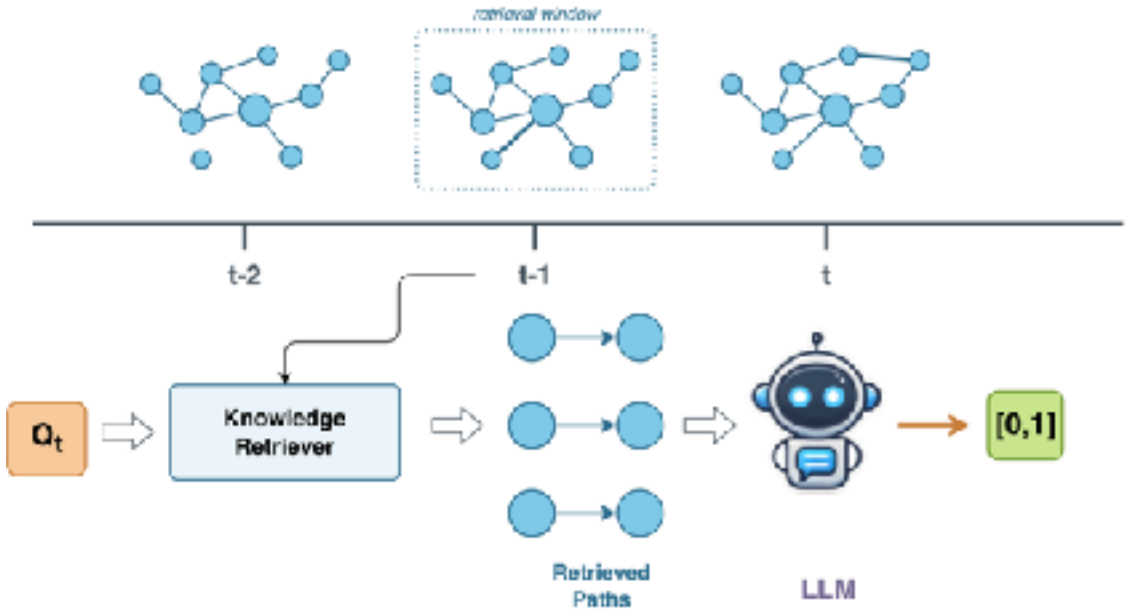
40–50% of Phase II efficacy failures trace to the target–disease hypothesis (Paul et al. 2010)

- **Reasoning is not forecasting**
  In general scientific domains, models reason well about mechanisms but rarely foresee which advances will happen

- **Forecasting or recall?**
  Only events after a model's training cutoff can separate the two; for novel target–disease pairs, this has not been tested

Bottleneck: Koller D., "Drug Discovery Has No Magic Wands" (2026); Paul S.M. et al., Nat. Rev. Drug Discov. 9:203–214 (2010). Forecasting: Wu et al., arXiv:2605.22681 (2026).

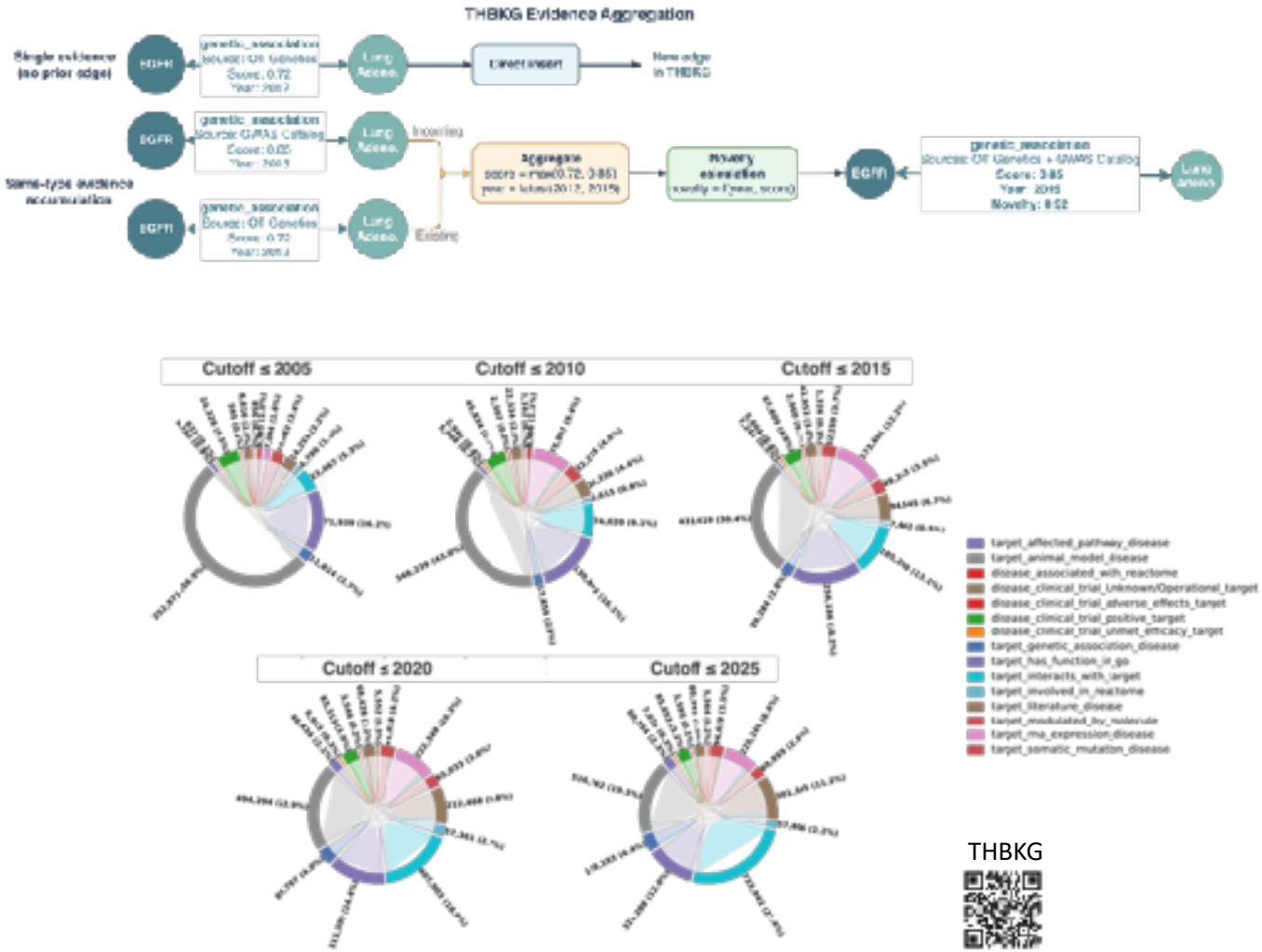
Retrospective validation: rank novel target–disease pairs that newly enter trials, using only evidence dated before that point



- **As-of query**
  $\mathcal{G}$[T] returns the graph as it stood before date T
- **Task**
  For a disease, rank candidate targets by whether they will newly enter a trial
- **Positives and negatives**
  First-in-history trial entries, 2021–2026, against 1:5 targets trialled for other diseases in the same therapeutic area
- **Cutoff overlap**
  Six open-weight LLMs, each scored against its own training cutoff; windows fall before and after it

Figure: Siu et al., TxCUSP (unpublished). Paths: K-Paths (Abdullahi et al., KDD 2025). Evidence: Open Targets (Ochoa et al., NAR 2023).

THBKG: a temporal biomedical knowledge graph oracle that reflects the target–disease landscape at each decision point



Paths retrieved with K-Paths (≤ 3 hops, top 8 per pair)

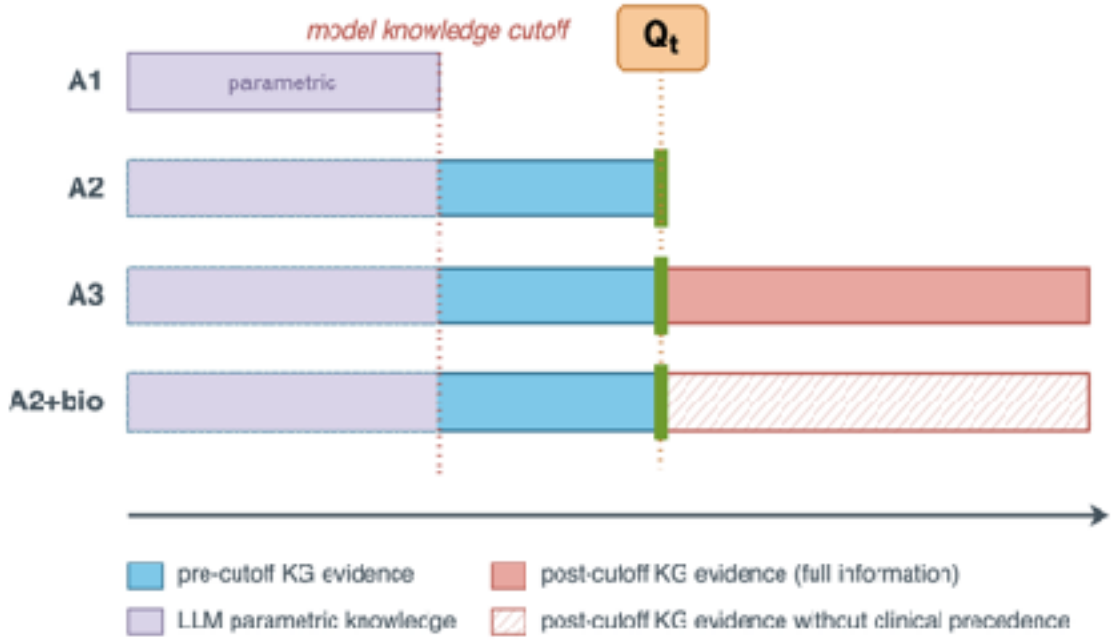
Prompt (A2 · dated paths)
System: All evidence presented below was available as of {cutoff date}. You have no access to information beyond this date.

```
TARGET: {symbol} ({Ensembl id})
DISEASE: {name} ({ontology id})
EVIDENCE PATHS connecting {symbol} to {name}:
Path 1: {gene} (target)
  --associated_with[score={score}]--> {node}
  (pathway) --involved_in[score={score}]-->
  {disease} (disease)
...
Will {symbol} enter a clinical trial as a
  therapeutic target for {name} after {window
  year}?
Answer with a single word: Yes or No.
```

Answer: one Yes/No token; its log-probability p_yes ranks each disease's candidates.

THBKG: Siu et al., arXiv:2608.05982 (2026). Path retrieval: K-Paths (Abdullahi et al., KDD 2025).

Four retrieval conditions separate knowledge, fresh biology and leakage



Condition	Retrieval graph	Post-cutoff edges	What it measures
A1	none (closed-book)	—	the **floor** — what the model already knows
A2	cutoff snapshot $\mathcal{G}$[C]	none	point-in-time retrieval: evidence at decision time
A2+bio	newest snapshot	trial edges removed	**label-masked oracle: fresh biology, no label**
A3	newest snapshot	all	**leaky oracle: full information, label leakage**

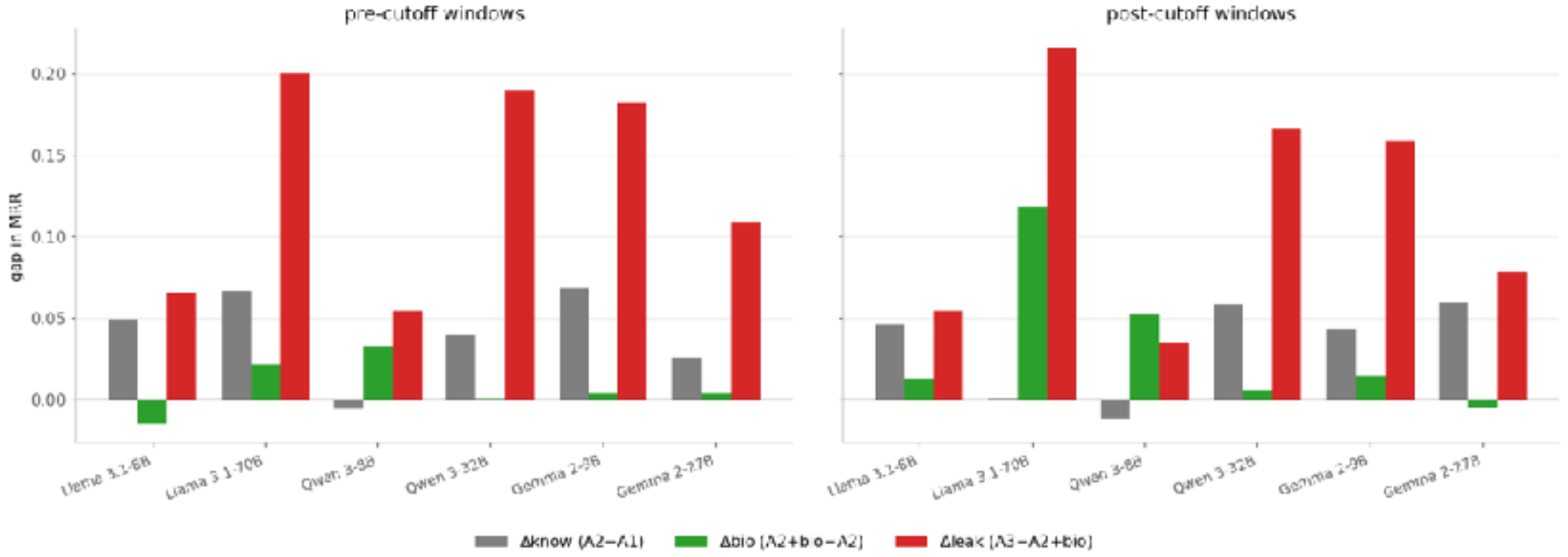
Closed-book models exceed chance; the leaky oracle's gain over this baseline requires explanation

					MRR		AUROC	
Model	Cutoff	n_{pre}	n_{post}	Cond.	pre	post	pre	post
Llama 3.1-8B	Dec 2023	925	447	A1	0.618	0.567	0.715	0.705
				A2	0.657	0.613	0.751	0.739
				A2+bio	0.652	0.626	0.723	0.712
				A3	0.718	0.681	0.798	0.778
Llama 3.1-70B	Dec 2023	925	447	A1	0.674	0.620	0.745	0.741
				A2	0.741	0.621	0.774	0.760
				A2+bio	0.762	0.739	0.792	0.798
				A3	0.963	0.955	0.975	0.971
Qwen 3-8B	Jun 2024*	1169	204	A1	0.614	0.592	0.731	0.730
				A2	0.608	0.581	0.721	0.705
				A2+bio	0.641	0.633	0.740	0.731
				A3	0.695	0.668	0.777	0.762
Qwen 3-32B	Jun 2024*	1164	204	A1	0.684	0.627	0.786	0.763
				A2	0.725	0.686	0.815	0.801
				A2+bio	0.726	0.692	0.805	0.801
				A3	0.916	0.858	0.955	0.928
Gemma 2-9B	Jun 2024*	1169	204	A1	0.623	0.606	0.725	0.737
				A2	0.692	0.649	0.770	0.745
				A2+bio	0.695	0.664	0.762	0.749
				A3	0.877	0.824	0.921	0.894
Gemma 2-27B	Jun 2024*	1169	204	A1	0.669	0.620	0.764	0.764
				A2	0.694	0.680	0.776	0.758
				A2+bio	0.698	0.675	0.776	0.767
				A3	0.807	0.754	0.878	0.855

- **A1 · closed-book**
  MRR ≈ 0.6 against a 0.38 chance baseline
- **A2 · point-in-time retrieval**
  Marginal gain over the closed-book baseline
- **A3 · leaky oracle (newest graph)**
  Highest scores, represent SOTA reported

*TxCUSP (unpublished). Disease-macro MRR / AUROC; chance MRR ≈ 0.38 at 1:5. *Approximate cutoff.*

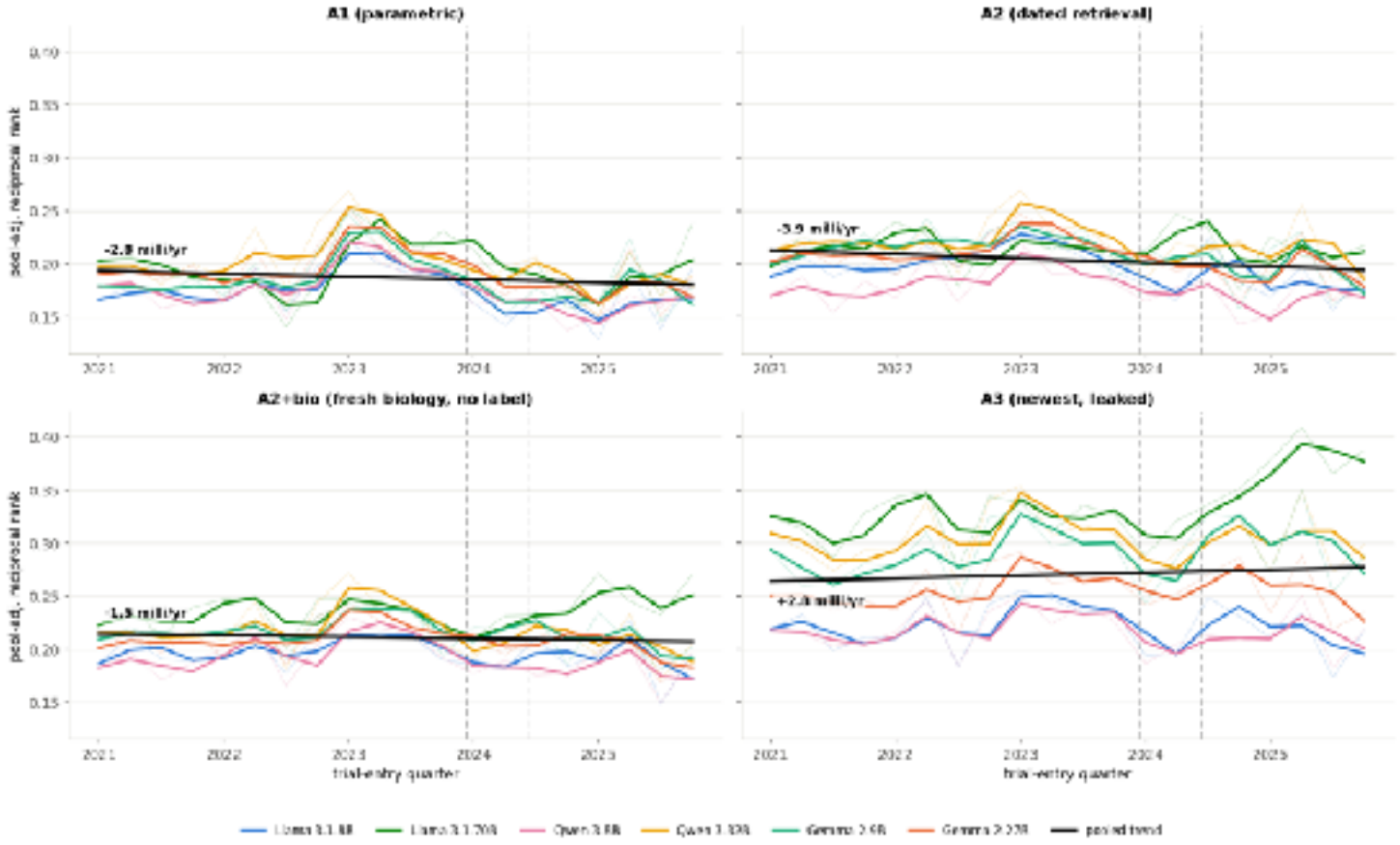
Point-in-time retrieval yields a modest gain; the leaky oracle's advantage derives from label leakage



Point-in-time retrieval: modest gain (3 of 6 models post-cutoff) · **Fresh biology:** no consistent gain (2 of 6)
Label leakage: principal source of the gain (12 of 12 cells)

TxCUSP (unpublished). Disease-macro MRR; AUROC gives the same ordering. Δknow = A2−A1 · Δbio = A2+bio−A2 · Δleak = A3−A2+bio.

Performance declines as pairs become more recent; only label leakage remains stable over time



Leakage-free conditions: decline with recency, with or without retrieval · **Post-cutoff biology:** attenuates the decline
Label leakage: time-invariant; the leaked edge encodes the label

TxCUSP (unpublished). Pool-adjusted reciprocal rank, two-quarter moving average; black = pooled trend.

LLM forecasting of novel pairs is real but modest; evaluation must time-bound evidence as well as labels

1. **LLMs show limited forecasting capability:** post-cutoff closed-book ranking exceeds chance
   a. **Point-in-time graph evidence adds a modest gain:** to be examined at all protein coding gene candidate pool

2. **Leakage-free performance declines with recency:**
   Evaluations should be stratified by knowledge cutoff or temporal window

3. **Gains from undated evidence reflect label leakage:** time-bound evidence, not only labels

4. **Clinical trial edges naturally carry registration dates:** time-bounding is a low-cost curation step

5. **A temporal evaluation harness for agentic discovery:**
   separates the evidence an LLM agent could have had at decision time from what arrived after

Next: RL on pre-cutoff outcomes, so models learn to reason over prior trial entries · evaluate beyond *whether*: *which mechanism* and *when*

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Acknowledgements

UKRI-BBSRC · AI for Drug Discovery Doctoral Training Programme, Queen Mary University of London · industry partner Recursion

Supervisors

Dr Arkaitz Zubiaga (QMUL EECS)
Dr Claudia Cabrera (William Harvey Research Institute, QMUL)
Dr Mani Mudaliar (Recursion)

AIDD programme leadership Prof Conrad Bessant · Prof Michael Barnes · Dr Mani Mudaliar · Emma Grant

Special thanks *to the AIDD cohort for early feedback and many useful conversations:*

Angela Abraham · Krishna Amin · Wei Dai · William Dee · Joy Efiannayi · Bernhard Finke · Stylianos Iliadis · **Yuan Liang** · Ivan Manoza · **Syed Zayyan Masud** · Constance Maurer · Caitlin Mellor · **Nasim Mohamed Ismail** · Marta Niedostatek · Martina Occhetta · **Aaron Wenteler** · **Yu-Yuan Yang** · **Ke Zhai**

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LLM forecasting of novel pairs is real but modest; evaluation must time-bound evidence as well as labels

1. **LLMs show limited forecasting capability:** post-cutoff closed-book ranking exceeds chance
   a. **Point-in-time graph evidence adds a modest gain:** to be examined at full phenome level

2. **Leakage-free performance declines with recency:**
   Evaluations should be stratified by knowledge cutoff or temporal window

3. **Gains from undated evidence reflect label leakage:** time-bound evidence, not only labels

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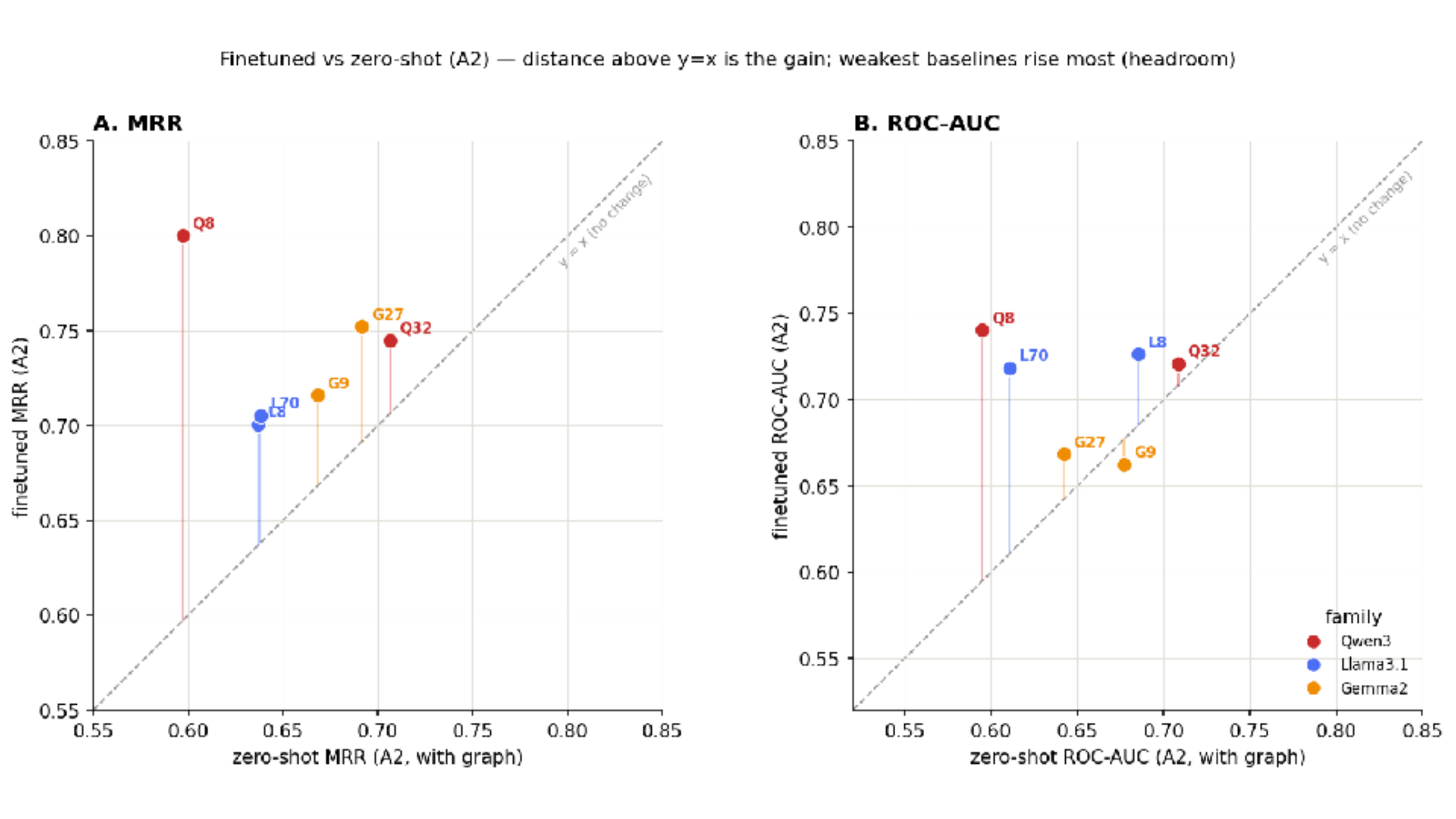
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Appendix


Finetuned vs zero-shot (A2) — distance above y=x is the gain; weakest baselines rise most (headroom)
A. MRR
B. ROC-AUC
finetuned MRR (A2)
zero-shot MRR (A2, with graph)
finetuned ROC-AUC (A2)
zero-shot ROC-AUC (A2, with graph)
y = x (no change)
Q8
L70
L8
G27
G9
Q32
family
Qwen3
Llama3.1
Gemma2
0.55
0.60
0.65
0.70
0.75
0.80
0.85

Full results: A3 leads for every model on both sides of the cutoff; A2 stays close to A1

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